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United States Patent	12418989
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Huang; Chun-Chieh et al.

Packaged circuit structure

Abstract

A package circuit structure includes a multilayer circuit board, an electronic component, and an insulating layer. The multilayer circuit board includes a metal portion and an opening. The opening is extending from a first side of the multilayer circuit board toward the second side of the multilayer circuit board facing the first side. A bottom of the opening is sealed by the metal portion. The electronic component is received in the opening and adhered to the metal portion. The electronic component is electrically connected to the multilayer circuit board and encapsulated in the opening by the insulating layer. A method for manufacturing the package circuit structure is also provided.

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Appl. No.: 18/123881

Filed: March 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230232537 A1	Jul. 20, 2023

Foreign Application Priority Data

CN	202110097582.X	Jan. 25, 2021
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Related U.S. Application Data

division parent-doc US 17218556 20210331 US 11683888 child-doc US 18123881

Publication Classification

Int. Cl.: **H05K1/18** (20060101); **H05K3/28** (20060101); **H05K3/46** (20060101)

U.S. Cl.:

CPC **H05K1/186** (20130101); **H05K3/284** (20130101); **H05K3/4697** (20130101); H05K1/183 (20130101); H05K2201/10553 (20130101); H05K2203/308 (20130101)

Field of Classification Search

CPC: H05K (1/186); H05K (1/182); H05K (1/183); H05K (1/184); H05K (1/185); H05K (3/4697); H05K (3/284)

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Background/Summary

FIELD

(1) The subject matter herein generally relates to a circuit board, particularly relates to a package circuit structure.

BACKGROUND

(2) With the development of the electronic products, sizes of the electronic products are becoming smaller. As an important element in the electronic products, size of the circuit board need to be smaller.

(3) Therefore, there is room for improvement within the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.

(2) FIG. 1 is a flowchart of an embodiment of a method for manufacturing a package circuit structure.

(3) FIG. 2 is a diagrammatic view of an embodiment of a multilayer circuit board.

(4) FIG. 3 is a flowchart of an embodiment of a method for manufacturing a multilayer circuit board.

(5) FIG. 4 is a diagrammatic view of an embodiment of a core board.

(6) FIG. 5 is a diagrammatic view of an embodiment of an inner circuit board formed by the core board of FIG. 4.

(7) FIG. 6 is a cross-sectional view showing a first outer circuit board and a second outer circuit board on the inner circuit board of FIG. 5.

(8) FIG. 7 is a cross-sectional view showing an opening formed on the multilayer circuit board of FIG. 2.

(9) FIG. 8 is a cross-sectional view showing two protective films on opposite sides of the multilayer circuit board of FIG. 2.

(10) FIG. 9 is a cross-sectional view showing an electronic component installed in the opening of FIG. 7.

(11) FIG. 10 is a cross-sectional view showing the electronic component of FIG. 9 packaged.

(12) FIG. 11 is a diagrammatic view of an embodiment of a package circuit structure.

DETAILED DESCRIPTION

(13) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

(14) The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.

(15) FIG. 1 illustrates a flowchart of a method for manufacturing a package circuit structure in accordance with an embodiment. The method is provided by way of embodiments, as there are a variety of ways to carry out the method. Each block shown in FIG. 1 represents one or more processes, methods, or subroutines carried out in the method. Furthermore, the illustrated order of blocks can be changed. Additional blocks may be added or fewer blocks may be utilized, without departing from this disclosure. The method can begin at block **201**.

(16) At block **201**, referring to FIG. 2, a multilayer circuit board **10** is provided. The multilayer circuit board **10** includes an opening area **101** extending from a first side **10a** of the multilayer circuit board **10** to a second side **10b** facing away from the first side **10a** along a thickness direction. The multilayer circuit board **10** further includes a metal portion **103** corresponding to a bottom of the opening area **101** to support the opening area **101**.

(17) In at least one embodiment, no wiring is provided in the opening area **101**. The multilayer circuit board **10** further includes two solder masks **18** on opposite sides of the multilayer circuit board **10** along the thickness direction. The opening area **101** is exposed from the solder mask **18** on the first side **10a**.

(18) In at least one embodiment, the multilayer circuit board **10** may be a circuit board with four wiring layers. FIG. 3 illustrates a flowchart of an embodiment of a method for manufacturing the multilayer circuit board **10**. The method can begin at block **401**.

(19) At block **401**, referring to FIG. 4, a core board **11** is provided. The core board **11** includes a first metal foil **113**, an inner dielectric layer **111**, and a second metal foil **115** stacked in that order along a thickness direction.

(20) At block **402**, referring to FIG. 5, an inner circuit board **11a** is formed by performing a circuit manufacturing technology on the core board **11**. The first metal foil **113** correspondingly forms a first inner wiring layer **113a**, the second metal foil **115** correspondingly forms a second inner wiring layer **115a**. The first inner wiring layer **113a** is electrically connected to the second inner wiring layer **115a**. The first inner wiring layer **113a** includes a metal portion **103**.

(21) In at least one embodiment, the second inner wiring layer **115a** may include a first gap **116** corresponding to the metal portion **103**. Preferably, a size of the first gap **116** may be greater than a size of the metal portion **103**, so that an orthographic projection of the metal portion **103** on the second inner wiring layer **115a** is located in the first gap **116**.

(22) At block **403**, referring to FIG. 6, a first outer circuit board **13** is built-up on the first inner wiring layer **113a**, and a second outer circuit board **15** is built-up on the second inner wiring layer **115a**. The first outer circuit board **13** includes a first outer dielectric layer **131** combined with the first inner wiring layer **113a** and a first outer wiring layer **133** formed on a side of the first outer dielectric layer **131** facing away from the inner circuit board **11a**. The second outer circuit board **15** includes a second outer dielectric layer **151** combined with the second inner wiring layer **115a** and a second outer wiring layer **153** formed on a side of the second outer dielectric layer **151** facing away from the inner circuit board **11a**.

(23) In at least one embodiment, the second outer wiring layer **153** may include a second gap **156** corresponding to the metal portion **103**. Preferably, a size of the second gap **156** may be greater than the size of the metal portion **103**, so that an orthographic projection of the metal portion **103** on the second outer wiring layer **153** is located in the second gap **156**.

(24) The first outer circuit board **13** and the second outer circuit board **15** may be electrically connected to the inner circuit board **11a**, respectively.

(25) At block **404**, referring to FIG. 2, two solder masks **18** are formed on a side of the first outer circuit board **13** facing away from the inner circuit board **11a** and a side of the second outer circuit board **15** facing away from the inner circuit board **11a**, respectively.

(26) In at least one embodiment, the solder mask **18** on the side of the second outer circuit board **15** facing away from the inner circuit board **11a** may include a third gap **183** corresponding to the metal portion **103**. Preferably, a size of the third gap **183** may be greater than the size of the metal portion **103**, so that an orthographic projection of the metal portion **103** on the solder mask **18** on the side of the second outer circuit board **15** facing away from the inner circuit board **11a** is located in the third gap **183**.

(27) In at least one embodiment, a portion of the second outer dielectric layer **151** corresponding to the metal portion **103** and a portion of the inner dielectric layer **111** corresponding to the metal portion **103** constitute the opening area **101**.

(28) At block **202**, referring to FIG. 7, the opening area **101** is removed to form an opening **101a**. The metal portion **103** seals the bottom of the opening **101a**.

(29) Preferably, a width of the opening **101a** may gradually decrease from the first side **10a** toward the second side **10b**, so that the metal portion **103** is beneficial to seal the bottom of the opening **101a**. In at least one embodiment, the width of the opening **101a** may be the same from the first side **10a** toward the second side **10b**.

(30) In at least one embodiment, the opening **101a** may be formed by mechanical cutting, laser cutting, plasma cutting or sandblasting.

(31) In at least one embodiment, before removing the opening area **101**, two protective films **50** (shown in FIG. 8) may be formed on the first side **10a** and the second side **10b** of the multilayer circuit board **10**. The opening area **101** is exposed from the protective film on the first side **10a**. The protective films **50** may reduce a risk of damage to other areas of the multilayer circuit board **10** when the opening area **101** is removed. After removing the opening area **101** to form the opening **101a**, the protective films **50** are removed.

(32) In at least one embodiment, the portion of the second outer dielectric layer **151** corresponding to the metal portion **103** and the portion of the inner dielectric layer **111** corresponding to the metal portion **103** are removed, thereby forming the opening **101a**.

(33) When no wiring or no solder mask is provided in the opening area **101**, a difference in materials when the opening area **101** is removed is reduced, which is beneficial to the formation of the opening **101a**. At the same time, the opening area **101** is supported by the metal portion **103**, which is beneficial to determine the cutting depth when the opening area **101** is removed.

(34) At block **203**, referring to FIG. 9, an electronic component **30** is installed in the opening **101a** and carried by the metal portion **103**. The electronic component **30** is electrically connected to the multilayer circuit board **10**.

(35) The electronic component **30** is carried by the metal portion **103**, which is beneficial to improve a bearing strength and also facilitate an adhesion of the electronic component **30** to the multilayer circuit board **10**.

(36) At block **204**, referring to FIG. 10, the electronic component **30** is packaged.

(37) In at least one embodiment, an insulating layer **60** is formed on the first side **10a** of the multilayer circuit board **10**. The insulating layer **60** fills the opening **101a** and covers the electronic component **30**, thereby encapsulating the electronic component **30**.

(38) The insulating layer **60** may be formed by injection molding.

(39) In at least one embodiment, a plurality of connecting pads **136** is formed on the second side **10b** of the multilayer circuit board **10**. The block **204** may further include forming a solder ball **40** on each connecting pad **136** for electrically connection with other components.

(40) Depending on the embodiment, certain of the steps of methods described may be removed, others may be added, and the sequence of steps may be altered. It is also to be understood that the description and the claims drawn to a method may include some indication in reference to sequential steps. However, the indication used is only to be viewed for identification purposes and not as a suggestion as to an order for the steps.

(41) FIG. 11 illustrates an embodiment of a package circuit structure **100**. The package circuit structure **100** includes a multilayer circuit board **10**, an electronic component **30**, and an insulating layer **60**. The multilayer circuit board **10** includes a metal portion **103** and an opening **101a**. The opening **101a** is extending from a first side **10a** of the multilayer circuit board **10** toward the second side **10b** of the multilayer circuit board **10** facing the first side **10a**. A bottom of the opening **101a** is sealed by the metal portion **103**. The electronic component **30** is received in the opening **101a** and adhered to the metal portion **103**. The electronic component **30** is electrically connected to the multilayer circuit board **10** and encapsulated in the opening **101a** by the insulating layer **60**.

(42) In at least one embodiment, a width of the opening **101a** may gradually decrease from the first side **10a** toward the second side **10b**, so that the metal portion **103** is beneficial to seal the bottom

of the opening **101a**. In at least one embodiment, the width of the opening **101a** may be the same from the first side **10a** toward the second side **10b**.

(43) The package circuit structure **100** may further include a plurality of solder balls **40**. Each solder ball **40** is formed on the second side **10b** of the multilayer circuit board **10** and electrically connected to the multilayer circuit board **10**.

(44) In at least one embodiment, the insulating layer **60** is formed on the first side **10a** of the multilayer circuit board **10**, fills in the opening **101a**, and covers the electronic component **30**.

(45) In at least one embodiment, the multilayer circuit board **10** may be a circuit board with four wiring layers. The multilayer circuit board **10** may include a dielectric layer **120**, a first wiring layer **121**, a second wiring layer **122**, a third wiring layer **123**, and a fourth wiring layer **124**. The first wiring layer **121**, the second wiring layer **122**, the third wiring layer **123**, and the fourth wiring layer **124** are stacked in that order and spaced with each other. The first wiring layer **121** and the fourth wiring layer **124** are respectively located on opposite sides of the dielectric layer **120**. The second wiring layer **122** and the third wiring layer **123** are embedded in the dielectric layer **120**. The multilayer circuit board **10** may further include two solder masks respectively formed on a side of the first wiring layer **121** facing away from the fourth wiring layer **124** and a side of the fourth wiring layer **124** facing away from the first wiring layer **121**.

(46) The second wiring layer **122** includes the metal portion **103**. The third wiring layer **123** may include a first gap **116** corresponding to the metal portion **103**. The fourth wiring **124** may include a second gap **156** corresponding to the metal portion **103**. The solder mask on the side of the fourth wiring layer **124** facing away from the first wiring layer **121** may include a third gap **183** corresponding to the metal portion **103**.

(47) The opening **101a** is recessed from a portion of the dielectric layer **120** exposed from the third gap **183** and the second gap **156** toward the second wiring layer **122** to expose the metal portion **103**.

(48) In the method for manufacturing the package circuit structure and the package circuit structure of the present disclosure, the opening is formed to receive the electronic component, thereby reducing an overall thickness of the package circuit structure. The bottom of the opening is sealed by the metal portion, which is beneficial to improve the bearing strength and facilitate the installation of the electronic component when the electronic component is installed.

(49) It is to be understood, even though information and advantages of the present embodiments have been set forth in the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only; changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present embodiments to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

Claims

1. A package circuit structure comprising: a multilayer circuit board, wherein the multilayer circuit board comprises a metal portion and an opening, the opening extends from a first side of the multilayer circuit board toward the second side of the multilayer circuit board facing the first side, a bottom of the opening is sealed by the metal portion; an electronic component, wherein the electronic component is received in the opening and adhered to the metal portion, the electronic component is electrically connected to the multilayer circuit board; and an insulating layer, wherein the insulating layer encapsulates the electronic component in the opening; wherein the multilayer circuit board further comprises a first wiring layer, a second wiring layer, a third wiring layer, and a fourth wiring layer stacked in that order; the second wiring layer comprises the metal portion, the third wiring layer comprises a first gap corresponding to the metal portion, the fourth wiring comprises a second gap corresponding to the metal portion; wherein a size of the first gap is greater

- than a size of the metal portion, and/or, a size of the second gap is greater than a size of the metal portion.
2. The package circuit structure of claim 1, wherein a width of the opening decreases from the first side of the multilayer circuit board toward the second side of the multilayer circuit board.
 3. The package circuit structure of claim 1, wherein a width of the opening is same from the first side toward the second side.
 4. The package circuit structure of claim 1, wherein the insulating layer is formed on the first side of the multilayer circuit board, fills in the opening, and covers the electronic component.
 5. The package circuit structure of claim 1, wherein the multilayer circuit board further comprises two solder masks respectively formed on the first wiring layer and the fourth wiring layer, the opening is exposed from one of the two solder masks on the fourth wiring layer.
 6. The package circuit structure of claim 5, wherein one of the two solder masks on the fourth wiring layer comprises a third gap corresponding to the metal portion.
 7. The package circuit structure of claim 6, wherein a size of the third gap is greater than a size of the metal portion.
 8. The package circuit structure of claim 6, wherein a plurality of connecting pads is formed on the second side of the multilayer circuit board, and a solder ball is formed on each of the plurality of connecting pads.
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